

## **WHO advice**

### **Surveillance:**

Based on the current situation and available information, WHO re-emphasizes the importance of strong surveillance by all Member States for acute respiratory infections, with the inclusion of MERS-CoV into the testing algorithm where warranted, and to carefully review any unusual patterns.

### **Clinical Management**

The incubation period is typically 2-15 days (median 5 days), although prolonged incubation periods have been reported in the immunocompromised. Although mild disease does occur, clinicians should be aware that symptoms may frequently progress rapidly non-specific signs of upper respiratory tract infection, cough and breathlessness, to respiratory failure and cardiovascular collapse.<sup>[3]</sup> MERS-CoV infection should be managed supportively with respiratory support titrated to the needs of the patient; there is a wide spectrum of severity, with many patients requiring mechanical ventilation.

The largest clinical trial in MERS compared a combination of lopinavir–ritonavir and interferon  $\beta$ -1b with placebo (95 patients).<sup>[4]</sup> Active treatment caused lower 90-day mortality in hospitalized patients with laboratory-confirmed MERS (90-day mortality of 48% and 29% respectively). Further analysis suggested a positive effect only in patients treated within 7 days of symptom onset. Although there is increasing use of corticosteroids for some respiratory conditions (specifically in COVID-19 and some other forms of pneumonia), their use in MERS-CoV is of uncertain benefit, and harms relating to their immunomodulatory effects may be significant; more data are needed. The use of convalescent plasma has not been proven, although has been used in a limited number of patients in a non-trial setting. While antibiotics have been used in severe disease to presumptively treat concurrent bacterial infection, there are no controlled data on efficacy. A retrospective analysis of 349 MERS patients examined macrolide antibiotic therapy. No difference in 90-day mortality was found in the 136 patients receiving macrolides compared with those who did not.<sup>[5]</sup>

### **Infection prevention and control:**

Human-to-human transmission of MERS-CoV in healthcare settings has been associated with delays in recognizing the early symptoms of MERS-CoV infection, slow triage of suspected cases and delays in implementing timely IPC measures. IPC measures are therefore critical to prevent the spread of MERS-CoV in healthcare facilities and onwards in the community. Healthcare workers should always apply standard precautions consistently with all patients and perform risk assessments at every interaction in healthcare settings to determine the necessary protection measures. For patients with suspected MERS-CoV infection that require hospitalization, place patient in an adequately ventilated single room away from other patient care areas. In addition to standard precautions. Droplet and contact precautions should be implemented when providing care to patients with symptoms of acute respiratory infection who are suspects of any respiratory disease, including probable or confirmed cases of MERS-CoV infection.<sup>[6,7]</sup>

Droplet and contact precautions should be maintained until the patient is no longer symptomatic (for at least 24 hours) and has two upper respiratory (URT) swabs (taken 24hrs apart) test negative in RT-PCR or according to local guidance. Additionally, airborne

precautions should be applied when performing aerosol generating procedures or in settings where aerosol generating procedures are conducted. Early identification, case management and prompt isolation of suspected respiratory infected patients and cases, quarantine of contacts, together with appropriate IPC measures in health care settings, including improving ventilation in enclosed spaces and public health awareness can prevent the spread of human-to-human transmission of MERS-CoV.

**Public health and social measures:**

MERS-CoV appears to cause more severe disease in people with underlying chronic medical conditions such as diabetes, renal failure, chronic lung disease, and immunosuppression. Therefore, people with these underlying medical conditions should avoid close contact with animals, particularly dromedaries, when visiting farms, markets, or barn areas where the virus may be circulating.

General hygiene measures, such as regular hand hygiene before and after touching animals or animal products and avoiding contact with sick animals, should be adhered to.

In addition, hygiene practices should be observed including the five keys to safer food should be followed when dealing with food items of camels; people should avoid drinking raw camel milk or camel urine or eating meat that has not been properly cooked.

WHO does not advise special screening at points of entry with regard to this event, nor does it currently recommend the application of any travel or trade restrictions.